

Beryllium-9

⁹Be (Be I), nuclear spin $I = 3/2$. Energy levels and transitions below 2 μm: 17 levels, 26 lines.

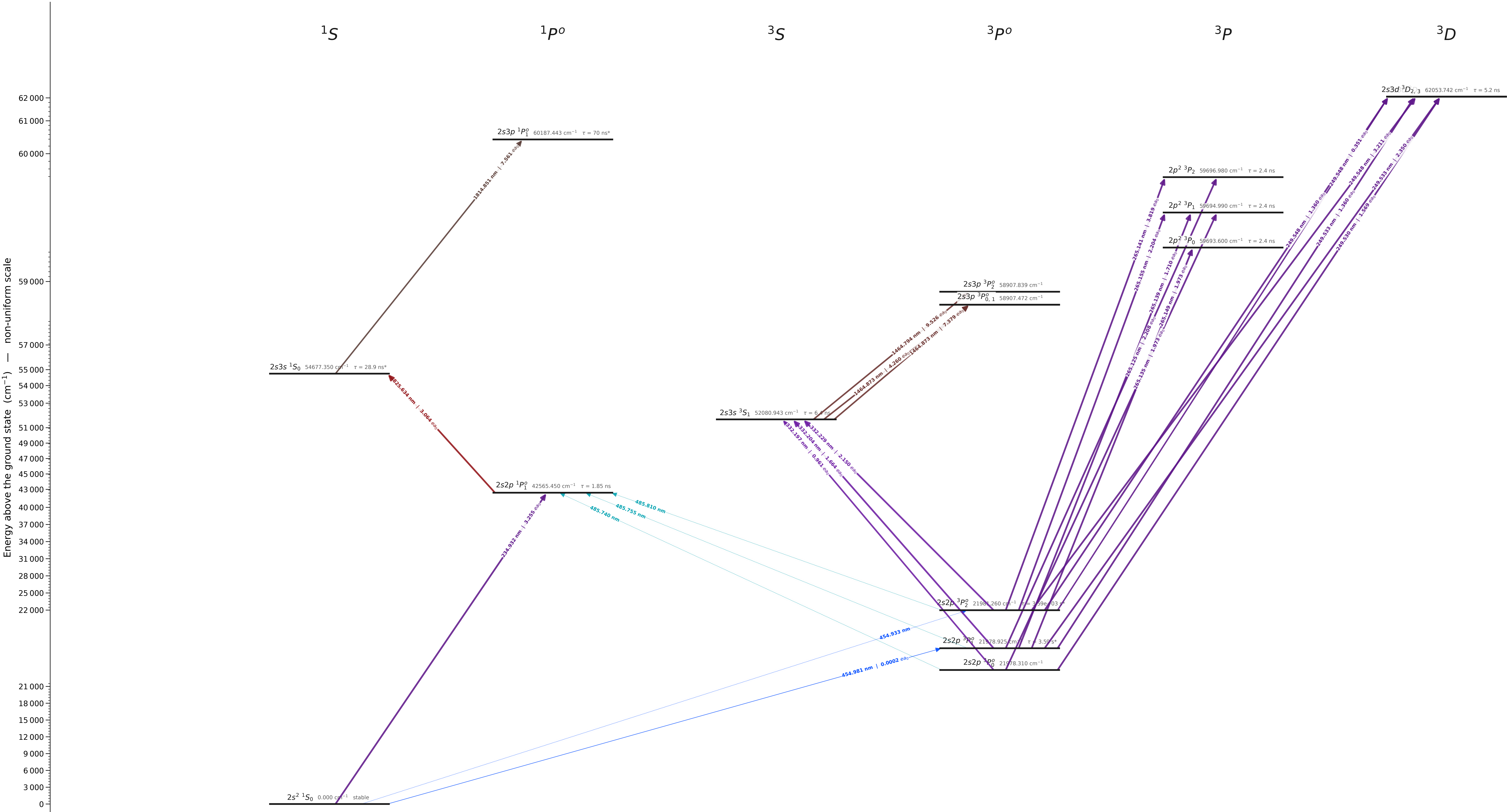
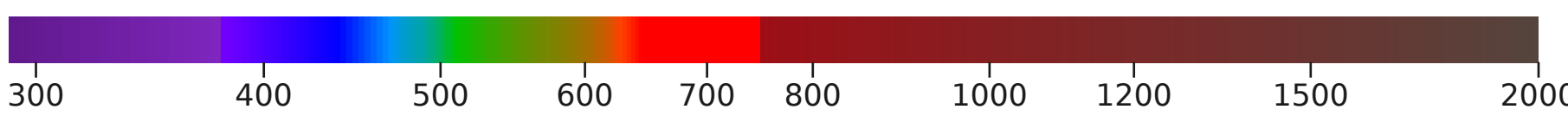
Be⁺ ionisation limit 75192.64 cm⁻¹ = 9.32270 eV (132.992 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U e U' $ (eao)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$2s^2\ ^1S_0 - 2s2p\ ^1P_1^o$	234.9323	234.8604	1276.08008(5)	3.255	5.520e+08	86.03	100	234.9 nm resonance line
$2s^2\ ^1S_0 - 2s2p\ ^3P_1^o$	454.9813	454.8538	658.91016	0.000	4.200e+01	4.43e-08	100	454.9 nm intercombination line
$2s2p\ ^1P_1^o - 2s3s\ ^1S_0$	825.6342	825.4073	363.10565(13)	3.064	3.380e+07	5.517	97.5	825.6 nm

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

⁹Be hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
2s2p $^3P_1^o$	-139.373(12)	-0.75(4)	F=1/2: +347.49	F=3/2: +140.13	F=5/2: -209.25	
2s2p $^3P_2^o$	-124.5368(17)	1.429(8)	F=1/2: +561.67	F=3/2: +373.61	F=5/2: +61.38	F=7/2: -373.25
2s2p $^1P_1^o$	-13.836	0.845	F=1/2: +35.65	F=3/2: +12.99	F=5/2: -20.54	
2s3s 3S_1	-332.250	-	F=1/2: +830.62	F=3/2: +332.25	F=5/2: -498.38	
2p 2 3P_1	18.101	0.734	F=1/2: -44.34	F=3/2: -18.84	F=5/2: +27.34	
2p 2 3P_2	3.386	-1.468	F=1/2: -16.52	F=3/2: -10.16	F=5/2: -0.78	F=7/2: +9.79
2s3p $^1P_1^o$	-6.704	0.427	F=1/2: +17.29	F=3/2: +6.28	F=5/2: -9.95	
2s3d 3D_1	150.600	0.016	F=1/2: -376.48	F=3/2: -150.62	F=5/2: +225.90	

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$2s^2\ ^1S_0 - 2s2p\ ^1P_1^o$	234.932	11-9	16015.29(3)

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$